

Shubham K. Singh

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EDUCATION

The National Institute of Engineering (NIE)

B.E. in Computer Science & Engineering | GPA: 8.9 (Current)

Aug. 2023 – Jun. 2027 (Expected)

Mysuru, India

- **Courses:** DBMS, OS, Networking, Machine Learning

Kendriya Vidyalaya COD Chheoki (CBSE)

Class 12: 94.6% | Class 10: 89.4%

Apr. 2020 – Mar. 2022

Prayagraj, India

PROJECTS

StockNet | Multi-VPC Trading Platform | AWS VPC, Lambda, API Gateway, EC2, RDS | [GitHub](#)

Jun. 2026

- Designed a **three-tier enterprise AWS architecture** with public/private subnet segmentation, VPC Peering across 3 VPCs, and Security Group-based network access controls.
- Deployed **Python Lambda functions** with IAM roles and API Gateway to expose secure HTTPS endpoints without managing application servers.
- Attached Lambda to the **VPC via ENIs** for private subnet access, enabling serverless communication with a FastAPI server on EC2 through internal IP addressing.
- Used **CloudWatch Logs** for request tracing and diagnostics; debugged connection refusals, subnet misconfigurations, and Security Group restrictions.

NeuroShield | AI Clinical Decision Support | React, Node.js, FastAPI, Supabase, LIME | [GitHub](#)

2025

- Built a **stroke risk prediction system** using a Balanced Random Forest trained on 1.2K+ patient records with **ROC-AUC of 0.88** and 85% recall, optimized to minimize false negatives in high-risk clinical scenarios.
- Integrated **LIME** to generate per-prediction feature importance scores and decision rules, making model transparent for clinical review.
- A **three-service architecture**: React frontend, Node.js/Express backend with role-based access (Patient, Doctor, Admin), and a **FastAPI ML microservice** serving inference and LIME explanations.

Mini OCR Pipeline | Product Feature Extraction | Python, PaddleOCR, ML Workflow

June 2025

- Built an **end-to-end OCR extraction pipeline** for an Amazon ML challenge, processing product images to extract unlabeled attributes (weight, dimensions, expiry dates).
- Implemented preprocessing to tackle noisy image data; engineered **custom post-processing logic** with regex, unit detection, and normalization.
- Benchmarked multiple OCR engines (PaddleOCR, EasyOCR, Tesseract) on **F1-score metrics**.

SKILLS

Backend Development : Node.js, Express.js, RabbitMQ, REST APIs, OAuth2.0, Docker

Databases : MongoDB, PostgreSQL

Languages : JavaScript, C++, Python

Machine Learning : OCR, Data preprocessing, Classification, Clustering, RAG

CS Fundamentals : Data Structures, OS, DBMS, Computer Networks, OOP

Tools : Git, GitHub, Vercel, Postman, n8n, Zod, GenAI

ACHIEVEMENTS & CERTIFICATIONS

- **Certifications:** **Hackerrank** - REST API(Intermediate), SQL(Intermediate).
- Real-world experience in product development for handling large number of users through **open-source contributions**, personal projects, and participating in global tech challenges.
- Organized learning sessions as part of **Anveshan (AIML Club)**, including ML bootcamps and tech talks.